



SHOPEZ PLATFORM SUITE

PROJECT DESIGN PHASE

Proposed Platform Solution Blueprint

System Components, Logic Workflows & High-Level Execution
Specifications

PROJECT NAME:	ShopEZ Premium E-Commerce Platform
REPOSITORY NAME:	VIP-C2-ShopEZ-E-commerce-Application
PREPARED BY:	P Pranay
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VERSION / STATUS:	v1.0.0 / Final Approved

1. Core Proposed Architecture

The proposed solution replaces static e-commerce operations with a highly dynamic, failure-tolerant architecture:

- Frontend Interface: Single Page Application built on React with Vite. Styled with vanilla CSS. Leverages CartContext and AuthContext to handle state changes dynamically.
- Backend Server: Express-Node API gateway handling security routing, file uploading using Multer, JSON data fallbacks, and JWT validation checks.
- Database Layer: MongoDB instance holding model collections. Equipped with a custom JSON reader fallback to ensure stable catalog rendering if database connectivity drops.

2. Process Integration Workflows

The system workflow guides the user seamlessly through navigation, cart operations, variant selection, and payment transactions, while checking block status parameters dynamically on each request.



